L7-8: Postulates of Quantum Mechanics, Linear Algebra

Vishvendra S. Poonia
Department of Electronics and Communication Engineering
Indian Institute of Technology Roorkee
vishvendra@ece.iitr.ac.in
vishvendra.punia@gmail.com

This session

Foundations of the “Quantum” Behavior

Postulates of Quantum Mechanics

The Mathematical Bedrock: Kets, Bras, Operators

Classical systems:

- State = list of numbers
- Example: Bit = 0 or 1
- Represented as scalar values

Why Do We Need a New Mathematical Language for Quantum Systems?

Quantum systems:

- State is not a definite value (SGz → SGz → same result) : (SGz → SGx → Splits again) : (SGz → SGx → SGz → Splits again!) **(The system must be described by something more than classical vectors)**
- **What mathematical structure naturally describes a system with exactly two possible outcomes? $\Longrightarrow$ A 2-dimensional complex vector space.**
- Needs a vector space representation.
- Spin is quantized. Measurement changes the state

$$|+\rangle_z = \frac{1}{\sqrt{2}}\left(|+\rangle_x + |-\rangle_x\right) \qquad |+\rangle_y = \frac{1}{\sqrt{2}}\left(|+\rangle_z + i|-\rangle_z\right)$$

This leads us to: Hilbert Space + Dirac Notation

Foundation: Postulates of Quantum Mechanics

- A system is represented by a wavefunction
  - Hilbert space
- Evolution is governed by Schrödinger equation
  - Hamiltonian
- Measurement
  - Hermitian operators
- **Superposition**
- **Entanglement**

Interference

Kets, Bras, Operators

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- **A Ket is a vector in a complex vector space. Notation:** $|\psi\rangle$
- Invented by Paul Dirac: "Bracket" → Bra + Ket

Example: Classical Bit vs Quantum Bit

- Classical bit: 0 or 1
- Quantum bit (qubit):
- These are basis vectors.

General Qubit State: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$

Where: $\alpha, \beta \in C$**.** $|\alpha|^2 + |\beta|^2 = 1$**.** This normalization ensures: Total probability = 1.

Kets, Bras, Operators

$$\langle\psi| = \begin{bmatrix}\alpha^* & \beta^*\end{bmatrix}$$

$$\lfloor\rho\rfloor$$

- A Ket is a vector in a complex vector space. Notation: $|\psi\rangle$
- Invented by Paul Dirac: "Bracket" → Bra + Ket

General Qubit State: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$**:** $\alpha, \beta \in C$**.** $|\alpha|^2 + |\beta|^2 = 1$**.**

A Bra is the dual vector of a ket. **Notation:** $\langle\psi|$**.** It is the conjugate transpose of the ket.

Why do we need both ket and bra?

- Because Hilbert Space is inner product vector space: **Inner Products (between a vector and its dual)**
- **Inner product** $\langle\phi|\psi\rangle$ **gives:**
  - Overlap between two states
  - **A complex number**
  - **Probability amplitude of projecting one state onto another**

Kets, Bras, Operators

Example: Inner product $\langle \phi \mid \psi \rangle$ gives:

- $\langle 0 \mid \psi \rangle = \alpha \Longrightarrow$ Probability of measuring 0: $\mid \langle 0 \mid \psi \rangle \mid^2 = \mid \alpha \mid^2$
- **This is the measurement rule.**

The Hilbert Space:

- **Quantum states live in:** Complex vector spaces with inner products → **Hilbert space**
- **Properties:** Linear, Complex-valued, Norm defined via inner product with Normalization: $\langle \psi \mid \psi \rangle = 1$

Operators in Quantum Mechanics:

- **Operators are:** Linear transformations acting on kets
- **Matrix representation:** $A \mid \psi \rangle$
- **Example:** Pauli-X Gate: $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ acts as: $X \mid 0 \rangle = \mid 1 \rangle$, $X \mid 1 \rangle = \mid 0 \rangle$

$\Longrightarrow$ It is the **quantum NOT gate**.

Thank you